



MISO DPP 2017 August Michigan Area Study Phase 3 Report

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(Revision 4-Final)

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1. Executive Summary

This report presents the results of a System Impact Study (SIS) performed to evaluate the interconnection of the generators in the DPP 2017 August Michigan Area Phase 3 (Michigan Area DPP-2017-August Phase 3). The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies:

- ATC
- MPower
- Consumers Energy
- PJM
- NIPSCO
- DTE
- Wolverine
- ITC

1.1. Project List

The original interconnection requests for DPP 2017 August Michigan Area had a total of 28 generation projects. Projects J812 & J820 & J925 withdrew prior to the start of Phase 1. Projects withdrawn after Phase 1 are J203 & J579 & J632 & J809 & J846 & J865 & J890 & J914 & J936. At Decision Point 1, J832 reduced from 100 MW to 91 MW, & J833 reduced from 100 MW to 50 MW, & J839 reduced from 100 MW to 58 MW. At Decision Point 2, J921 and J931 withdrew. Therefore, there are 14 generation projects with a combined nameplate rating of 3681 MW (ERIS) & 3624 MW (NRIS). The detailed list is shown below in Table 1, and the Phase 3 study was kicked off on April 24 2019.

Table 1: List of DPP 2017 August Michigan Area Phase 3 Projects

Project #	Transmission Owner	County	State	Service Type	POI Name	Summer MW	Winter MW	Fuel
J701	ITC Transmission	Tuscola	MI	NRIS	Rapson - Grassmere - Bauer 345kV Line Tap (Kirk 345 kV Sub)	49	49	Wind
J793	ITC Transmission	St. Clair	MI	NRIS	Belle River - Fitz #2 345 kV Line Tap	1175	1209	Combined Cycle
J794	METC	Gratiot	MI	NRIS	J589 138kV Substation (Regal - Summerton 138kV Line)	12.5	12.5	Wind
J796	METC	Shiawassee	MI	NRIS	Goss 138kV Substation	239	239	Solar
J799	ITC Transmission	Washtenaw	MI	NRIS	Milan - Pioneer 120kV Line Tap	189	189	Solar
J806	METC	Calhoun	MI	NRIS	Rice Creek - Vrooman 138kV Line Tap	149	149	Solar
J832	METC	Cheboygan	MI	NRIS	Port Calcite - Riggsville 138kV Line	91	91	Solar
J833	ITC Transmission	Washtenaw	MI	NRIS	Milan-Pioneer 120kV line tap	50	50	Solar
J839	ITC Transmission	Lapeer	MI	NRIS	Otsego Tap - Otsego 120 kV Line Tap	58	58	Solar
J841	METC	Saginaw	MI	NRIS	Orr RD 138 kV Substation	702	725	Combined Cycle
J857	METC	Calhoun	MI	NRIS	Verona - Foundry 138 kV Line Tap	124.95	124.95	Solar
J875	METC	Lenawee	MI	NRIS	Par - Raisin Road 138kV Line	150	150	Solar
J906	METC	Muskegon	MI	NRIS	Kenowa - Ludington #2 345 kV Line Tap	100	100	Wind
J937	METC	Calhoun	MI	NRIS	Argenta - Tompkins 345kV Line Tap	507	535	Combined Cycle



1.2. Total Network Upgrades

The cost allocation of Network Upgrades for the projects in the DPP 2017 August Michigan Phase 3 reflects responsibilities for mitigating system impacts. The total cost of network upgrades is listed in Table 2 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies.

Table 2: Total Cost of Network Upgrades for DPP 2017 August Michigan Phase 3 Projects

Project	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)	M4 (\$)
	Thermal	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned					
a	B	C	d	e	f	g	H	i	j = b+c+d+f+g+i	\$4,000/MW	(10% of (j) from Phase I) - M2	(20% of (j) from Phase II) -M2 - M3
J701	\$-	\$-	\$-	\$-	\$176,344	\$-	\$-	\$-	\$176,344	\$196,000	\$-	\$-
J793	\$-	\$-	\$-	\$1,215,555	\$1,763,642	\$9,614,000	\$657,000	\$-	\$11,377,642	\$5,500,000	\$-	\$-
J794	\$-	\$-	\$-	\$200,000	\$98,452	\$-	\$-	\$-	\$98,452	\$50,000	\$608,971	\$-
J796	\$-	\$-	\$-	\$178,426	\$993,491	\$2,700,000	\$210,000	\$-	\$3,693,491	\$956,000	\$-	\$-
J799	\$-	\$-	\$-	\$166,099	\$-	\$7,310,000	\$200,000	\$-	\$7,310,000	\$756,000	\$291,813	\$791,032
J806	\$3,222,500	\$2,501,916	\$-	\$108,620	\$-	\$6,561,000	\$356,000	\$-	\$12,285,416	\$596,000	\$1,352,291	\$1,699,788
J832	\$-	\$1,000,000	\$-	\$-	\$225,147	\$6,809,000	\$421,000	\$485,606	\$8,519,754	\$400,000	\$2,361,537	\$-
J833	\$-	\$-	\$-	\$-	\$-	\$10,290,000	\$180,000	\$-	\$10,290,000	\$400,000	\$407,050	\$468,262
J839	\$-	\$-	\$-	\$-	\$82,891	\$20,360,000	\$200,000	\$-	\$20,442,891	\$400,000	\$691,614	\$894,425
J841	\$740,000	\$-	\$-	\$529,220	\$4,212,744	\$1,570,000	\$720,000	\$2,619,496	\$9,142,240	\$2,900,000	\$-	\$-
J857	\$1,857,500	\$2,098,084	\$-	\$206,380	\$-	\$12,730,000	\$420,000	\$-	\$16,685,584	\$500,000	\$1,537,214	\$-
J875	\$480,000	\$-	\$-	\$750,000	\$-	\$41,955,000	\$225,000	\$-	\$42,435,000	\$600,000	\$142,532	\$438,213
J906	\$-	\$-	\$-	\$-	\$117,289	\$11,007,000	\$306,000	\$-	\$11,124,289	\$400,000	\$673,922	\$1,001,419
J937	\$1,000,000	\$-	\$-	\$-	\$-	\$10,408,000	\$471,000	\$-	\$11,408,000	\$2,028,000	\$1,323,939	\$-
Total	\$7,300,000	\$5,600,000	\$-	\$3,354,300	\$7,670,000	\$141,314,000	\$4,366,000	\$3,105,102	\$164,989,102	\$15,682,000	\$9,390,883	\$5,293,138

Analyses performed demonstrate the following transmission facilities are required to reliably interconnect this group of generators to the transmission system. Energy Resource Interconnection Service (ERIS) Network Upgrades and Network Resource Interconnection Service (NRIS) Network Upgrades are shown in Table 3. Shared Network Upgrades are shown in Table 4.



Table 3: ERIS & NRIS Upgrades (Planning level cost estimates)

Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
Sag remediation on J806 POI-Denso Jackson 138 kV	METC	J806, J857		\$2,880,000
A10-Conditionality-MTEP19 Target A Project #12003 resolves the issue on J833 POI-Pioneer 120 kV	ITCT	J833, J799		A10
Sag remediation on J875 POI-Parr Road 138 kV	METC	J875		\$240,000
Sag remediation on J875 POI-Raisin 138 kV	METC	J875		\$240,000
Replace Terminal Equipment (MTEP ID 13814)+increase rating (MTEP ID 15878) on Monroe-Lallendorf 345 kV	ATSI/ITCT	J701, J793, J796, J799, J833, J839, J841, J875	J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937	A10
Replace 138kV Terminal Equipment at Tompkins.	METC	J937		\$1,000,000
A10-Conditionality-MTEP19 Target A Project #15798 increases the limit on Cornell-Goss 138 kV to above the observed rating. Pending MTEP 19 approval	METC	J796	J796	A10
Rebuild 2 miles of 336 ACSR to 1431 ACSR on Denso Jackson-Vrooman 138 kV	METC	J806, J857		\$2,200,000
Reconductor 0.38 miles of 954 ACSR on HSC-Orr Road 138 kV	METC	J841		\$500,000
Sag remediation on Lawndale-Murphy 138 kV	METC	J841		\$240,000
A10-Conditionality-MTEP19 Project #16028 Caniff-Saturn 120 kV	ITCT	J793		A10
A10-Conditionality-MTEP19 Project #17164 will reduce the flow below the rating on Jewel to St. Clair #1 345 kV line.	ITCT	J793		A10
A10-Conditionality-MTEP19 Project #17164 will reduce the flow below the rating on Belle River to St. Clair 345 kV line.	ITCT	J793		A10
A10-Conditionality-MTEP19 Target A project #16023 increases Rating-replace the wavetrap for the 19NWEST-19CTL12 120 kV constraint	ITCT	J793		A10
A10-Conditionality-MTEP19 Target A project #16023 increases Rating-Reconductor the overhead for the 19L-N-NW-19CTL3 120 kV constraint	ITCT	J793		A10
Sag remediation+ Replace span of 2156 ACSR on Thetford-Jewell 345 kV	METC/ITCT		J701, J794, J796, J832, J841, J906	\$1,000,000
Sag remediation on Bennigton- Sncklj 138 kV	METC		J796	\$250,000
Sag remediation on Gleaner-Hackett 138 kV	METC		J794, J841	\$720,000
Sag remediation on Hackett-Laundra 138 kV	METC		J794, J841	\$1,200,000
Sag remediation and terminal equipment upgrade on Pontiac-Placid 345 kV	ITCT		J701, J793, J794, J796, J832, J839, J841, J906	\$1,000,000
Sag remediation and terminal equipment upgrade on Pontiac-Wixom 345 kV	ITCT		J701, J793, J794, J796, J832, J839, J841, J906	\$1,000,000
Sag remediation and terminal equipment upgrade on Bauer-Pontiac 345 kV	ITCT		J701, J794, J796, J832, J841, J906	\$1,500,000
A10-Conditionality-MTEP19 Project #15852 will increase Rate B on Vanat J to Sncklj 138 kV above identified flow	METC		J796	A10

Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
A10-Conditionality-MTEP19 Target A Project #15939 will increase Rate B on Mercer- Cement 138 kV above identified flow	METC		J806, J857, J937	A10
A10-Conditionality- Replace bus work at Beveridge Substation as previously scheduled - December 2018 completion	METC		J796	A10
A10-Conditionality-DPP 2017 FEB Network upgrade increases sag limit on Gleaner-Bullock 138kV	METC		J794	A10
Sag remediation on Placid-Wixom 345 kV	ITCT		J701, J793, J794, J796, J832, J839, J841, J906	\$1,000,000

Table 4: Shared Network Upgrades (Planning level cost estimates)

Shared Network Upgrade	TO	Higher queued projects associated with SNU	Study projects associated with SNU	Cost of solution (\$)
Emmet-Livingston 138 kV	WPSC	J392	J832	\$ 600,000
Vestaburg 138-69 kV Transformer	WPSC	J589	J841	\$ 3,093,500

Note:

- (1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- (2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.
- (3) Analysis performed shows there are four projects for Shared Network Upgrade cost allocation.

2. Model Development and Study Assumptions

2.1. Base Case Models

The following summer peak and shoulder peak year 2022 models were developed based on Phase 3 study models. The origin of the DPP 2017 August Michigan models is the MTEP 2017 models with the Bench Cases including all pre-queued projects and associated network upgrades, while the Study Cases contain all of the interconnection requests in DPP 2017 August Michigan Phase 3, in addition to all the facilities in the Bench Cases.

- Bench Cases
 - Bench-DPP2017AUG-MI2022SH70IN_LP-Steady_v33_190605ph3.raw
 - Bench-DPP2017AUG-MI2022SH85IN_LG-Steady_v33_190605ph3.raw
 - Bench-DPP2017AUG-MI2022SH85Z_LG-Steady_v33_190605ph3.raw
 - Bench-DPP2017AUG-MI2022SPK_IN_LG-Steady_v33_190605ph3.raw
 - Bench-DPP2017AUG-MI2022SPK_Z_LG-Steady_v33_190605ph3.raw
- Study Cases
 - Study-DPP2017AUG-MI2022SH70IN_LP-Steady_v33_190605ph3.raw
 - Study-DPP2017AUG-MI2022SH85IN_LG-Steady_v33_190605ph3.raw
 - Study-DPP2017AUG-MI2022SH85Z_LG-Steady_v33_190605ph3.raw
 - Study-DPP2017AUG-MI2022SPK_IN_LG-Steady_v33_190605ph3.raw
 - Study-DPP2017AUG-MI2022SPK_Z_LG-Steady_v33_190605ph3.raw

2.2. Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

2.3. Contingencies

The following contingencies were considered in the steady state analysis:

- (1) NERC Category P0 (system intact -- no contingencies)
- (2) NERC Category P1 contingencies
 - a. Single element outages
 - b. Multiple element NERC Category P1 contingencies
- (3) NERC Category P2-P7 contingencies
- (4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

2.4. Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2017 August Michigan generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Models. Analyses used PSS®E version 33.11.0 and TARA version 1802b.

2.5. Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or

- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and have DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as Injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner FERC filed Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the Deliverability algorithm, to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

3. Thermal Analysis

The thermal analysis results for 2022 Summer Peak and Summer Shoulder scenarios show generator projects J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937 causing constraints. The details pertaining to the thermal analysis can be found in Appendix A – MISO ERIS Analysis (CEII).

3.1. J701

The thermal analysis identified the generator to contribute to the following constraint:

1. Monroe-Lallendorf 345 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. MTEP Projects #13814 and 15878 will reduce the flow on this line below the required rating. If the MTEP projects are not approved, there will be 5% of the cost assigned to J701.

3.2. J793

The thermal analysis identified the generator to contribute to the following constraints:

1. Monroe-Lallendorf 345 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. MTEP Projects #13814 and 15878 will reduce the flow on this line below the required rating. If the MTEP projects are not approved, there will be 49% of the cost assigned to J793.

2. Caniff-Saturn 120 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. MTEP Project #16028 will reduce the flow on this line below the required rating. If the MTEP project is not approved, there will be 100% of the cost assigned to J793.

3. Jewel-St. Clair 345 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. MTEP19 Project #17164 will reduce the flow on this line below the required rating. Applicable rate B should be 1348 MVA. If the MTEP project is not approved, there will be 100% of the cost



assigned to J793.

4. Belle River-St. Clair 345 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. MTEP19 Project #17164 will reduce the flow on this line below the required rating. Applicable rate B should be 1348 MVA. If the MTEP project is not approved, there will be 100% of the cost assigned to J793.

5. Northwest to CTL 120 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. MTEP19 Target A project #16023, Lincoln-Northeast-Northwest 120 kV Rebuild (only the wave-trap replacement), increases rating of the line to above the required rating. If the MTEP project is not approved, there will be 100% of the cost assigned to J793.

6. Lincoln-Northeast-Northwest to CTL 120 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. MTEP19 Target A project #16023, Lincoln-Northeast-Northwest 120 kV Rebuild (only the overhead section re-conductor), increases rating of the line to above the required rating. If the MTEP project is not approved, there will be 100% of the cost assigned to J793.

3.3. J794

No mitigations were found to be required for this generator.

3.4. J796

The thermal analysis identified the generator to contribute to the following constraints:

1. Monroe-Lallendorf 345 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. MTEP Projects #13814 and 15878 will reduce the flow on this line below the required rating. If the MTEP project are not approved, there will be 7% of the cost assigned to J796.

2. Goss-Cornell 138 kV

Per MISO cost allocation rules, the project receives no cost allocation for upgrades required to mitigate the above listed constraint. MTEP19 Target A Project #15798 increases the limit on the line and alleviates the constraint. The project is allocated 0% of the cost. The upgrade is pending MTEP 19 approval and J796 is conditional on the MTEP project being in service. If the MTEP project is not approved, there will be 100% of the cost assigned to J796.

3.5. J799

The thermal analysis identified the generator to contribute to the following constraints:

1. Monroe-Lallendorf 345 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. MTEP Projects #13814 and 15878 will reduce the flow on this line below the required rating. If the MTEP projects are not approved, there will be 8% of the cost assigned to J799.

2. J833 POI-Pioneer 120 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. MTEP Project #12003 will reduce the flow on this line below the required rating. If the MTEP project is not approved, there will be 79% of the cost assigned to J799.

3.6. J806

The thermal analysis identified the generator to contribute to the following constraints:

1. J806 POI-Denso Jackson 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is sag remediation. The cost estimate is \$2.88 million, with the project allocated 63% of the cost.

2. Denso Jackson- Vrooman 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild 2 miles of 336 ACSR to 1431 ACSR. Resulting limit is a CETO limit of 259 MVA. The cost estimate is \$2.2 million, with the project allocated 63% of the cost.

3.7. J832

No mitigations were found to be required for this generator.

3.8. J833

The thermal analysis identified the generator to contribute to the following constraints:

3. Monroe-Lallendorf 345 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. MTEP Projects #13814 and 15878 will reduce the flow on this line below the required rating. If the MTEP projects are not approved, there will be 2% of the cost assigned to J833.

4. J833 POI-Pioneer 120 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. MTEP Project #12003 will reduce the flow on this line below the required rating. If the MTEP project is not approved, there will be 21% of the cost assigned to J833.

3.9. J839

The thermal analysis identified the generator to contribute to the following constraints:

1. Monroe-Lallendorf 345 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. MTEP Projects #13814 and 15878 will reduce the flow on this line below the required rating. If the MTEP projects are not approved, there will be 2% of the cost assigned to J839.

3.10. J841

The thermal analysis identified the generator to contribute to the following constraints:

1. Monroe-Lallendorf 345 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the



above listed constraint. MTEP Projects #13814 and 15878 will reduce the flow on this line below the required rating. If the MTEP projects are not approved, there will be 22% of the cost assigned to J841.

2. HSC-Orr Road 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to reconductor 0.38 miles of 954 ACSR on HSC-Orr Road 138 kV. In addition to that, sag remediation is also required. The cost estimate is \$0.5 million, with the project allocated 100% of the cost.

3. Lawndale-Murphy 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is sag remediation. The cost estimate is \$0.24 million, with the project allocated 100% of the cost

3.11. J857

The thermal analysis identified the generator to contribute to the following constraints:

1. J806 POI-Denso Jackson 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is sag remediation. The cost estimate is \$2.88 million, with the project allocated 37% of the cost.

2. Denso Jackson- Vrooman 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild 2 miles of 336 ACSR to 1431 ACSR. Resulting limit is a CETO limit of 259 MVA. The cost estimate is \$2.2 million, with the project allocated 37% of the cost.

3.12. J875

The thermal analysis identified the generator to contribute to the following constraints:

1. J875 POI-Parr Road 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is sag remediation. The cost estimate is \$.24 million, with the project allocated 100% of the cost.

2. J875 POI-Raisin 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is sag remediation. The cost estimate is \$.24 million, with the project allocated 100% of the cost.

3. Monroe-Lallendorf 345 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. MTEP Projects #13814 and 15878 will reduce the flow on this line below the required rating. If the MTEP projects are not approved, there will be 4% of the cost assigned to J875.



3.13. J906

No mitigations were found to be required for this generator.

3.14. J937

The thermal analysis identified the generator to contribute to the following constraints:

1. Tompkins 345-138-13.8 kV xfmr #2

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to replace terminal equipment at Tumpkins. The cost estimate is \$1 million, with the project allocated 100% of the cost.

4. Voltage Analysis

The voltage analysis results for 2022 Summer Peak and Summer Shoulder scenarios show that the study generators do not cause any voltage violations. The details pertaining to the voltage analysis can be found in Appendix A – Appendix A – MISO ERIIS Analysis (CEII) .

5. Stability Analysis

.Stability analysis for projects J701, J794, J796, J799, J806, J832, J833, J839, J857, J875, and J906, were performed by Siemens. The Stability analysis show that generator projects J806, J832, and J857 have some adverse impact on the system Details pertaining to the stability analysis can be found in Appendix B Siemens Stability Analysis Report (CEII).

Stability analysis for projects J793, J841, and J937 were performed by ITC. No adverse impact was identified for these projects. Details pertaining to the stability analysis can be found in Appendix C ITC Stability Analysis Report (CEII).

5.1. J701

Stability Network Upgrades are not required for interconnection of the J701 generating facility.

5.2. J793

Stability Network Upgrades are not required for interconnection of the J793 generating facility

5.3. J794

Stability Network Upgrades are not required for interconnection of the J794 generating facility.

5.4. J796

Stability Network Upgrades are not required for interconnection of the J796 generating facility.

5.5. J799

Stability Network Upgrades are not required for interconnection of the J799 generating facility.

5.6. J806

For interconnection of the J806 generating facility, ITC will cut in the current Marshall-Blackstone 138 kV line into the J857 substation. The cost estimate is \$4.6 million, with the project allocated 54% of the cost

5.7. J832

For interconnection of the J832 generating facility, ITC will install dual pilot relaying on the McGulpin-Riggsville 138 kV line and the Riggsville-Livingston 138 kV line. The cost estimate is \$1 million, with the project allocated 100% of the cost.

5.8. J833



Stability Network Upgrades are not required for interconnection of the J833 generating facility.

5.9. J839

Stability Network Upgrades are not required for interconnection of the J839 generating facility.

5.10. J841

Stability Network Upgrades are not required for interconnection of the J841 generating facility.

5.11. J857

For interconnection of the J806 generating facility, ITC will cut in the current Marshall-Blackstone 138 kV line into the J857 substation. The cost estimate is \$4.6 million, with the project allocated 46% of the cost.

5.12. J875

Stability Network Upgrades are not required for interconnection of the J875 generating facility.

5.13. J906

Stability Network Upgrades are not required for interconnection of the J906 generating facility.

5.14. J937

Stability Network Upgrades are not required for interconnection of the J937 generating facility.

6. Short Circuit Analysis

ITC performed Short Circuit analysis utilizing CAPE software. Single line to ground faults and three phase faults were evaluated for pre- and post-project cases (similar to Bench and Study models). Short circuit study of DPP 2017 Aug MI Phase 3 indicates that no project adversely impacts any circuit breaker capability. The details pertaining to the short circuit analysis are presented in Appendix D – ITC Short Circuit Report (CEII).

7. Affected System Impact Study

The details pertaining to the PJM, CE, DTE, and ATC Affected Systems studies are in, Appendix E – PJM Affected Systems Study Report (CEII), Appendix F – CE Affected Systems Study Report (CEII), Appendix G – DTE Affected Systems Study Report (CEII), and Appendix I – ATC Affected Systems Study Report (CEII).

7.1. J701

No mitigations were found to be required for this generator.

7.2. J793

Per PJM analysis, this generator was found to contribute to the following constraints:

1. Lallendorf - bay 345 kV line

Per PJM cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to reconductor 2.0 miles of the line. The cost estimate is \$2.0893 million, with the project allocated 58.18% of the cost.

7.3. J794

Per CE analysis, this generator was found to contribute to the following constraints:

1. Grover T – Alma Prod J 46 kV Line for contingency outage of Alma 138 kV Bus

Per CE cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild 0.52 miles of the 46 kV line. The cost estimate is \$0.2

million, with the project allocated 100% of the cost.

7.4. J796

Per PJM analysis, this generator was found to contribute to the following constraints:

1. Lallendorf - bay 345 kV line

Per PJM cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to reconductor 2.0 miles of the line. The cost estimate is \$2.0893 million, with the project allocated 8.54% of the cost.

7.5. J799

Per PJM analysis, this generator was found to contribute to the following constraints:

1. Lallendorf - bay 345 kV line

Per PJM cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to reconductor 2.0 miles of the line. The cost estimate is \$2.0893 million, with the project allocated 7.95% of the cost.

7.6. J806

Per CE analysis, this generator was found to contribute to the following constraints:

1. Chauncey J–Calhoun 46 kV Line for contingency outage of Calhoun-Guardian J 46 kV Line

Per CE cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild 0.34 miles of ChaunceyJ-Calhoun 46 kV Line. The cost estimate is \$0.2 million, with the project allocated 54.3% of the cost.

7.7. J832

No mitigations were found to be required for this generator.

7.8. J833

No mitigations were found to be required for this generator.

7.9. J839

No mitigations were found to be required for this generator.

7.10. J841

Per PJM analysis, this generator was found to contribute to the following constraints:

1. Lallendorf - bay 345 kV line

Per PJM cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to reconductor 2.0 miles of the line. The cost estimate is \$2.0893 million, with the project allocated 25.33% of the cost.

7.11. J857

Per CE analysis, this generator was found to contribute to the following constraints:

1. Chauncey J–Calhoun 46 kV / Calhoun–Guardian J 46 kV Line

Per CE cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild 0.34 miles of ChaunceyJ-Calhoun 46 kV Line. The cost estimate is \$0.2 million, with the project allocated 45.7% of the cost.

2. An electrical island of existing generation at the Convis Substation and 138 kV line connected load



is formed for outage of J857 Switching Station to J758 Switching Station 138 kV Line

Per CE cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to move DTT from the Foundry Substation to the J857 Switching Station. The cost estimate is \$0.115 million, with the project allocated 100% of the cost.

7.12. J875

Per CE analysis, additional distribution upgrade work required by Consumers Energy to meet METC J875 protection requirements. The cost estimate is \$0.75 million, with the project allocated 100% of the cost.

7.13. J906

No mitigations were found to be required for this generator.

7.14. J937

No mitigations were found to be required for this generator.

8. Deliverability Analysis

8.1. Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource (ER) project or proceed with the system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

8.2. Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator's output is not deliverable.

8.3. Deliverability Study Results

The limiting constraints (mon-con pairs) seen in the deliverability analysis for the 2022 Summer case (MI2022SPK_Z_LG) are summarized in Appendix H – MISO Deliverability Analysis (CEII) .

8.4. J701

This generator is found to be fully deliverable for 0 MW.

This generator is determined to be fully deliverable for 49 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 5 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 5: NRIS Results for J701

J701 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and case assumptions)				0 MW (0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Thetford-Jewell 345 kV	0	0.1617	No		J701, J794, J796, J832, J841, J906	3.9%	\$1,000,000
Bauer-Pontiac 345 kV	0	0.2046	No		J701, J794, J796, J832, J841, J906	4.9%	\$1,500,000
Pontiac-Placid 345 kV	0	0.2217	No		J701, J793, J794, J796, J832, J839, J841, J906	2.1%	\$1,000,000
Pontiac-Wixom 345 kV	0	0.2121	No		J701, J793, J794, J796, J832, J839, J841, J906	2.1%	\$1,000,000
Monroe-Lallendorf 345 kV	41.2	0.2439	Yes	J701, J793, J796, J799, J833, J839, J841, J875	J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937	0%	A10 Conditionality on MTEP project #s 13814 & 15878
Placid-Wixom 345kV	49	0.1617	No		J701, J793, J794, J796, J832, J839, J841, J906	2.1%	\$1,000,000

8.5. J793

This generator is found to be fully deliverable for 425.7 MW without any network upgrades.

This generator is determined to be fully deliverable for 1175 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 6 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 6: NRIS Results for J793

J793 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and case assumptions)				425.7 MW (36.2%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Pontiac-Placid 345 kV	539.3	0.2566	No		J701, J793, J794, J796, J832, J839, J841, J906	59.2%	\$1,000,000
Pontiac-Wixom 345 kV	713.78	0.2442	No		J701, J793, J794, J796, J832, J839, J841, J906	58.9%	\$1,000,000
Monroe-Lallendorf 345 kV	1113.92	0.2798	Yes	J701, J793, J796, J799, J833, J839, J841, J875	J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937	0%	A10 Conditionality on MTEP project #s 13814 & 15878
Placid-Wixom 345kV	1175	0.2293	No		J701, J793, J794, J796, J832, J839, J841, J906	58.2%	\$1,000,000

8.6. J794

This generator is found to be fully deliverable for 0 MW without any network upgrades.

This generator is determined to be fully deliverable for 12.5, contingent upon the system upgrades and

contingent facilities identified in the NRIS analysis. Table 7 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 7: NRIS Results for J794

J794 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERS and case assumptions)				0 MW (0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Gleaner-Hackett 138 kV	0	0.1007	No		J794, J841	3.2%	\$720,000
Thetford-Jewell 345 kV	0.64	0.152	No		J701, J794, J796, J832, J841, J906	0.9%	\$1,000,000
Hackett-Laundra 138 kV	3.25	0.1007	No		J794, J841	3.3%	\$1,200,000
Bauer-Pontiac 345 kV	4.53	0.157	No		J701, J794, J796, J832, J841, J906	1.0%	\$1,500,000
Pontiac-Placid 345 kV	5.74	0.146	No		J701, J793, J794, J796, J832, J839, J841, J906	0.4%	\$1,000,000
Pontiac-Wixom 345 kV	7.59	0.1406	No		J701, J793, J794, J796, J832, J839, J841, J906	0.4%	\$1,000,000
Monroe-Lallendorf 345 kV	11.85	0.1896	Yes	J701, J793, J796, J799, J833, J839, J841, J875	J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937	0%	A10 Conditionality on MTEP project #s 13814 & 15878
Placid-Wixom 345kV	12.5	0.1362	No		J701, J793, J794, J796, J832, J839, J841, J906	0.4%	\$1,000,000

8.7. J796

This generator is found to be fully deliverable for 0 MW without any network upgrades.

This generator is determined to be fully deliverable for 239 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 8 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 8: NRIS Results for J796

J796 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERS and case assumptions)				0 MW (0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Thetford-Jewell 345 kV	36.46	0.1844	No		J701, J794, J796, J832, J841, J906	21.7%	\$1,000,000
Bennigton- Snckj 138 kV	62.16	0.0868	No		J796	100.0%	\$250,000
Bauer-Pontiac 345 kV	86.59	0.1676	No		J701, J794, J796, J832, J841, J906	19.5%	\$1,500,000
Pontiac-Placid 345 kV	89.93	0.1629	No		J701, J793, J794, J796, J832, J839, J841, J906	7.6%	\$1,000,000
Vanat J-Snckj 138 kV	109.7	0.0868	No		J796	0%	A10 Conditional on MTEP19

							Target A Project #15852
Pontiac-Wixom 345 kV	145.19	0.1576	No		J701, J793, J794, J796, J832, J839, J841, J906	7.7%	1,000,000
Monroe-Lallendorf 345 kV	205.77	0.2145	Yes	J701, J793, J796, J799, J833, J839, J841, J875	J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937	0%	A10 Conditionality on MTEP project #s 13814 & 15878
Goss-Cornell 138 kV	226.58	0.2352	Yes	J796	J796	0%	\$0 A10 Conditionality on MTEP19 Target A Project #15798
Placid-Wixom 345kV	239	0.1547	No		J701, J793, J794, J796, J832, J839, J841, J906	8.0%	\$1,000,000

8.8. J799

This generator is found to be fully deliverable for 114.81 MW without any network upgrades.

This generator is determined to be fully deliverable for 189 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 9 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 9: NRIS Results for J799

J799 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIIS and case assumptions)				114.81 MW (60.7%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe-Lallendorf 345 kV	189	0.2866	Yes	J701, J793, J796, J799, J833, J839, J841, J875	J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937	0%	A10 Conditionality on MTEP project #s 13814 & 15878

8.9. J806

This generator is found to be fully deliverable for 90.51 MW without any network upgrades.

This generator is determined to be fully deliverable for 149 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 10 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 10: NRIS Results for J806

J806 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIIS and case assumptions)				90.51 MW (60.7%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)

Monroe-Lallendorf 345 kV	146.42	0.1664	Yes	J701, J793, J796, J799, J833, J839, J841, J875	J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937	0%	A10 Conditionality on MTEP project #s 13814 & 15878
Mercer- Cement 138 kV	149	0.0779	No		J806, J857, J937	0%	A10 Conditional on MTEP19 Target A Project #15939

8.10. J832

This generator is found to be fully deliverable for 0 MW without any network upgrades.

This generator is determined to be fully deliverable for 91 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 11 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 11: NRIS Results for J832

J832 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and case assumptions)				0 MW (0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Thetford-Jewell 345 kV	23.67	0.1356	No		J701, J794, J796, J832, J841, J906	6.1%	\$1,000,000
Bauer-Pontiac 345 kV	32.97	0.1397	No		J701, J794, J796, J832, J841, J906	6.2%	\$1,500,000
Pontiac-Placid 345 kV	41.77	0.1318	No		J701, J793, J794, J796, J832, J839, J841, J906	2.4%	\$1,000,000
Pontiac-Wixom 345 kV	55.28	0.1269	No		J701, J793, J794, J796, J832, J839, J841, J906	2.4%	\$1,000,000
Monroe-Lallendorf 345 kV	86.27	0.1762	Yes	J701, J793, J796, J799, J833, J839, J841, J875	J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937	0%	A10 Conditionality on MTEP project #s 13814 & 15878
Placid-Wixom 345kV	91	0.1231	No		J701, J793, J794, J796, J832, J839, J841, J906	2.4%	\$1,000,000

8.11. J833

This generator is found to be fully deliverable for 30.37 MW without any network upgrades.

This generator is determined to be fully deliverable for 50 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 12 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 12: NRIS Results for J833

J833 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and case assumptions)				30.37 MW (60.7%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe-Lallendorf 345 kV	50	0.287	Yes	J701, J793, J796, J799, J833, J839, J841, J875	J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937	0%	A10 Conditionality on MTEP project #s 13814 & 15878

8.12. J839

This generator is found to be fully deliverable for 21.01 MW without any network upgrades.

This generator is determined to be fully deliverable for 58 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 13 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 13: NRIS Results for J839

J839 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and case assumptions)				21.01 MW (36.2%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Pontiac-Placid 345 kV	26.62	0.24	No		J701, J793, J794, J796, J832, J839, J841, J906	2.7%	\$1,000,000
Pontiac-Wixom 345 kV	35.23	0.231	No		J701, J793, J794, J796, J832, J839, J841, J906	2.8%	\$1,000,000
Monroe-Lallendorf 345 kV	54.98	0.2715	Yes	J701, J793, J796, J799, J833, J839, J841, J875	J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937	0%	A10 Conditionality on MTEP project #s 13814 & 15878
Placid-Wixom 345kV	58	0.2237	No		J701, J793, J794, J796, J832, J839, J841, J906	2.8%	\$1,000,000

8.13. J841

This generator is found to be fully deliverable for 0 MW without any network upgrades.

This generator is determined to be fully deliverable for 702 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 14 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 14: NRIS Results for J841

J841 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and case assumptions)				0 MW (0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Thetford-Jewell 345 kV	0	0.1851	No		J701, J794, J796, J832, J841, J906	64.0%	\$1,000,000
Gleaner-Hackett 138 kV	143.57	0.0521	No		J794, J841	96.7%	\$720,000
Hackett-Laundra 138 kV	182.38	0.0521	No		J794, J841	96.7%	\$1,200,000
Bauer-Pontiac 345 kV	254.33	0.191	No		J701, J794, J796, J832, J841, J906	65.3%	\$1,500,000
Pontiac-Placid 345 kV	322.2	0.1768	No		J701, J793, J794, J796, J832, J839, J841, J906	24.4%	\$1,000,000
Pontiac-Wixom 345 kV	426.44	0.1698	No		J701, J793, J794, J796, J832, J839, J841, J906	24.5%	\$1,000,000
Monroe-Lallendorf 345 kV	665.5	0.2133	Yes	J701, J793, J796, J799, J833, J839, J841, J875	J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937	0%	A10 Conditionality on MTEP project #s 13814 & 15878
Placid-Wixom 345kV	702	0.1851	No		J701, J793, J794, J796, J832, J839, J841, J906	24.8%	\$1,000,000

8.14. J857

This generator is found to be fully deliverable for 75.9 MW without any network upgrades.

This generator is determined to be fully deliverable for 124.95 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 15 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 15: NRIS Results for J857

J857 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS and case assumptions)				75.9 MW (60.7%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe-Lallendorf 345 kV	122.79	0.1592	Yes	J701, J793, J796, J799, J833, J839, J841, J875	J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937	0%	A10 Conditionality on MTEP project #s 13814 & 15878
Mercer- Cement 138 kV	124.95	0.0753	No		J806, J857, J937	0%	A10 Conditional on MTEP19 Target A Project #15939

8.15. J875

This generator is found to be fully deliverable for 91.12 MW without any network upgrades.

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and



contingent facilities identified in the NRIS analysis. Table 16 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 16: NRIS Results for J875

J875 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERS and case assumptions)				91.12 MW (60.7%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe-Lallendorf 345 kV	150	0.2374	Yes	J701, J793, J796, J799, J833, J839, J841, J875	J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937	0%	A10 Conditionality on MTEP project #s 13814 & 15878

8.16. J906

This generator is found to be fully deliverable for 0 MW without any network upgrades.

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 17 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 17: NRIS Results for J906

J906 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERS and case assumptions)				0 MW (0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Thetford-Jewell 345 kV	26.01	0.0674	No		J701, J794, J796, J832, J841, J906	3.3%	\$1,000,000
Bauer-Pontiac 345 kV	36.23	0.0652	No		J701, J794, J796, J832, J841, J906	3.2%	\$1,500,000
Pontiac-Placid 345 kV	45.9	0.0605	No		J701, J793, J794, J796, J832, J839, J841, J906	1.2%	\$1,000,000
Pontiac-Wixom 345 kV	60.75	0.0587	No		J701, J793, J794, J796, J832, J839, J841, J906	1.2%	\$1,000,000
Monroe-Lallendorf 345 kV	94.8	0.1151	Yes	J701, J793, J796, J799, J833, J839, J841, J875	J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937	0%	A10 Conditionality on MTEP project #s 13814 & 15878
Placid-Wixom 345kV	100	0.0579	No		J701, J793, J794, J796, J832, J839, J841, J906	1.3%	\$1,000,000

8.17. J937

This generator is found to be fully deliverable for 335.99 MW without any network upgrades.

This generator is determined to be fully deliverable for 535 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis.

Table 18 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 18: NRIS Results for J937

J937 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIIS and case assumptions)				335.99 MW (62.8%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Monroe-Lallendorf 345 kV	526.24	0.1503	Yes	J701, J793, J796, J799, J833, J839, J841, J875	J701, J793, J794, J796, J799, J806, J832, J833, J839, J841, J857, J875, J906, J937	0%	A10 Conditionality on MTEP project #s 13814 & 15878
Mercer- Cement 138 kV	535	0.0553	No		J806, J857, J937	0%	A10 Conditional on MTEP19 Target A Project #15939

9. Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) test for Network Upgrades driven by higher queued interconnection projects was performed for this System Impact Study.

The maximum MW impacts and Shared Network Upgrade (SNU) cost allocations appear in Table 19.

Table 19: Maximum MW Impact and SNU Cost Allocations

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility (\$)
Emmet-Livingston 138 kV	DPP 2015 Feb MI	J392	4.0	\$ 600,000	\$114,394
	DPP 2017 Aug MI	J832	17.2		\$485,606
Vestaburg 138-69 kV Transformer	DPP 2016 Aug MI	J589	1.7	\$ 3,093,500	\$474,004
	DPP 2017 Aug MI	J841	9.5		\$2,619,496

10. Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

10.1. Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade identified in System Impact Study was provided by the corresponding transmission owning company.

10.2. Cost Allocation Methodology

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.

Cost Allocation Methodology for Thermal Constraints

1. With all Study Group generation projects dispatched in the Post Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any network upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

$$\text{Cost of NU} = \frac{\text{Project A cost portion of NU} \times \text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contrution on constraint)}}$$

6. The total NU costs for each project are calculated if more than one NU is required.



Appendix A – Appendix A – MISO ERIS Analysis (CEII)

Appendix B – Siemens Stability Report (CEII)

Appendix C – ITC Stability Report (CEII)

Appendix D – ITC Short Circuit Report (CEII)

Appendix E – PJM Affected Systems Study Report (CEII)

Appendix F – CE Affected Systems Study Report (CEII)

Appendix G – DTE Affected Systems Study Report (CEII)

Appendix H – MISO Deliverability Analysis (CEII)

Appendix I – ATC Affected Systems Study Report (CEII)